

EDUCATION

University of California San Diego, San Diego, CA

B.S. in Nanoengineering

GPA: 3.811/4.00

Expected Graduation: June 2026

CURRENT RESEARCH AND PROJECTS

Broad-band Single-mode Waveguides, Fainman Lab, UC San Diego

01 2025 – Present

- Working with two graduate students to make a chip-scale white light emitter of a single mode. Not authorized to talk about details.

Waveguide Simulations, Fainman Lab, UC San Diego

09 2024 – Present

- Responsible for using the open-source software package FEniCSx to solve for the normal modes of dielectric waveguides and resonators. Familiar with mesh generation and the mathematics behind the Ritz and Galerkin finite element methods.

Optical Gravimeter, SPIE Optics Club, UC San Diego

09 2024 – Present

- Responsible for designing the chassis, alignment system, and stabilization system of the optical gravimeter. Came up with the idea to precisely control the center of gravity (COG) of the hanging gravimeter chassis and to stabilize the COG equilibrium position by suspending the device in a viscous liquid.

Capstone Project "Enhanced Thermocouple", UC San Diego

09 2024 – Present

- Project Manager. A conductive graphene monolith will be enhanced via the intercalation of copper microspheres or carbon nano-tubes (CNTs). Achievement of in-plane alignment of CNTs is a major interest of the project.

Molecular Systems Design

09 2024 – Present

- Doing self-guided research to create a program which converts receptor diagrams into synthesizable molecules. Fields currently being explored include quantum chemistry, quantum electrodynamics, supramolecular chemistry, organic chemistry, ab-initio simulations, molecular dynamics simulations, molecular biology, and host-guest binding.

PAST PROJECTS AND ROLES

Eagle Scout Project, Troop 730, Morgan Hill

03 2020 – 09 2021

- Coordinated with Alameda Homeless Shelter client to gather specifications and gauge resident desires. Computer modeled, generated blueprints, and guided troop to build a rollable, lockable, caged garden bed for the Alameda Homeless Shelter, San Jose. Contraption was successfully delivered and reassembled on-site. Time spent: 150 hours. Documents available on request.

First Robotics, Valley Christian, San Jose

09 2019 – 06 2022

- Computer-aided Design Lead for Team 3256. Worked with other team members to design a compact "intestinal" robotic feeding system to transfer balls to a robotic flywheel launcher. The robot was a success, making it to worlds in 2021.

Blender, Node Form

12 2022 – 03 2024

- Used Python to program a plug-in for Blender which allows the user to input mathematical transformations into a node-based interface. Blender then creates an animation by generating mesh-geometry and transforming any selected objects. Time spent: 300 hours. Code available at github.com/FyDininno.

TECHNICAL SKILLS

- Computers: Java, Python, Linux systems, Bash, Multi-booting, Github, CAD (Blender, Autodesk Inventor, Onshape)
- Manufacturing: Filament Printing, Resin Printing, Laser Cutting, Lathing, and other manual and power tools

RELEVANT COURSEWORK

- Linear Systems, Fourier Analysis, Electromagnetic Optics, Optical Information Processing, Materials Science, Density Functional Theory, Crystallography, Organic Chemistry, Manifold Integration, Heat Transfer, Circuit Analysis, Molecular Orbital Theory, Quantum Mechanics, Abstract Algebra, Photolithography, Microfabrication.

BOOK KNOWLEDGE AND PERSONAL INTERESTS

- Science: Supramolecular Chemistry, Clifford Algebra formulation of Relativistic Electrodynamics, Clifford Algebra representation of Euclidean Groups
- Student Orgs: UCSD Club Swim, UCSD Tritonthenics (Calisthenics), UCSD SPIE Optics Club